Benchmarking LLMs via Uncertainty Quantification

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Objective

Problem

Current LLM leaderboards (HuggingFace, OpenCompass . . .) rank models by **accuracy only**.

Two models with *equal accuracy* may differ greatly in **uncertainty** – yet benchmarks treat them identically.

Goal

Integrate **uncertainty quantification** into LLM benchmarking via **conformal prediction** – a statistically rigorous, distribution-free, model-agnostic method.

Methodology – Conformal Prediction

Coverage Guarantee

For test instance X_t with true label Y_t, CP outputs a **prediction set** $\mathcal{C}(X_t) \subseteq \mathcal{Y}$ such that:

$$P\big(Y_t \in \mathcal{C}(X_t)\big) \;\geq\; 1 - \alpha$$

α = user-specified error rate (e.g. $\alpha = 0.1 \;\Rightarrow$ **90 %** coverage guarantee).

Uncertainty Measure

Set Size (SS) = average $|\mathcal{C}(X_t)|$ over all test instances.

Small SS $\Rightarrow$ **confident model** Large SS $\Rightarrow$ **uncertain model**

Methodology – Score Functions

LAC – Least Ambiguous Classifier

$$s(X, Y) = 1 - f(X)_Y$$

Uses softmax score of the true label only.

$\Rightarrow$ tends to smaller sets.

APS – Adaptive Prediction Sets

$$s(X, Y) = \sum_{\substack{Y' \in \mathcal{Y}: \\ f(X)_{Y'} \geq f(X)_Y}} f(X)_{Y'}$$

Accumulates ranked softmax scores.

$\Rightarrow$ better on hard instances.

Final metrics averaged over **LAC** *and* **APS**.

Algorithm – CP for LLMs

Algorithm 1: Conformal Prediction for LLMs Input: $\mathcal{M}$, $\mathcal{D}$, α Output: Acc, SS, CR, UAcc

for each prompt strategy $p \in \{\text{base, shared, task}\}$ **do** // *3 strategies*
  for each ICL setting $k \in \{\text{0-shot, few-shot, CoT}\}$ **do** // *3 ICL modes*
    $P(X) \leftarrow \mathrm{Softmax}(\mathrm{logits}_{\mathcal{M}}(X_{p,k}))$ **for all** $X \in \mathcal{D}$ // *query LLM*
    $\mathcal{D} \rightarrow \mathcal{D}_{\mathrm{cal}}$ (50%) $+ \mathcal{D}_{\mathrm{test}}$ (50%) // *random split*
    for each $s \in \{\text{LAC, APS}\}$ **do** // *2 score fns*
      $s_i \leftarrow s(X_c^{(i)}, Y_c^{(i)})$ **for all** $i \in \mathcal{D}_{\mathrm{cal}}$ // *nonconformity scores*
      $\hat{q} \leftarrow \mathrm{Quantile}\left(\{s_i\}, \frac{\lceil (n+1)(1-\alpha) \rceil}{n}\right)$ // *adaptive threshold*
      $\mathcal{C}(X_t) \leftarrow \{Y' \in \mathcal{Y} : s(X_t, Y') \leq \hat{q}\}$ **for all** $X_t \in \mathcal{D}_{\mathrm{test}}$ // *prediction sets*
      if $\mathcal{C}(X_t) = \emptyset$ **then keep** $\arg\max_{Y'} P(X_t)_{Y'}$
return avg over p, k, s: $\mathrm{Acc},\ \mathrm{SS},\ \mathrm{CR},\ \mathrm{UAcc} = \frac{\mathrm{Acc}}{\mathrm{SS}}\sqrt{|\mathcal{Y}|}$

Methodology – Evaluation Metrics

Metric	Formula	Meaning
Acc	$\frac{1}{\lvert\mathcal{D}_{\text{test}}\rvert}\sum \mathbf{1}[Y_p = Y_t]$	Prediction accuracy
SS	$\frac{1}{\lvert\mathcal{D}_{\text{test}}\rvert}\sum \lvert\mathcal{C}(X_t)\rvert$	Uncertainty (↑ more uncertain)
CR	$\frac{1}{\lvert\mathcal{D}_{\text{test}}\rvert}\sum \mathbf{1}[Y_t \in \mathcal{C}(X_t)]$	Coverage ($\geq 1-\alpha$ guaranteed)

New: Uncertainty-aware Accuracy (UAcc)

$$\mathrm{UAcc} = \frac{\mathrm{Acc}}{\mathrm{SS}}\sqrt{|\mathcal{Y}|}, \qquad |\mathcal{Y}| = 6$$

Rewards models that are both **accurate** and **certain**. Can **reverse rankings** vs. Acc alone.

Results – All Base Models

Source: Table 1 – mean of LAC & APS, avg. over 5 tasks

LLM	Acc (%) ↑	SS ↓	UAcc ↑
Yi-34B	81.46	1.93	121.09
Qwen-72B	78.05	2.06	109.96
Qwen-14B	74.00	2.17	102.28
Llama-2-70B	72.24	2.16	94.62
DeepSeek-67B	71.66	2.15	91.07
Yi-6B	68.97	2.36	83.53
Mistral-7B	64.14	2.43	70.84
Llama-2-13B	60.42	2.56	61.07
Qwen-7B	59.87	2.63	64.34
InternLM-7B	49.31	3.41	40.83
Llama-2-7B	46.53	3.09	38.71
DeepSeek-7B	45.87	3.13	36.80
Qwen-1.8B	42.34	3.38	32.95

Results – Three Key Findings

Finding 1: Higher Accuracy $\neq$ Lower Uncertainty

InternLM-7B outperforms MPT-7B by **+19.34 %** Acc on DRS, yet shows **higher** SS (3.63 vs. 3.59). Rankings by Acc and SS diverge across *all* tasks.

Finding 2: Larger $\neq$ More Certain

Qwen 1.8B$\rightarrow$14B: Acc and certainty both improve.
Qwen 14B$\rightarrow$72B: Acc rises, but SS **increases** on RC & DRS – non-monotonic.

Finding 3: Instruction Finetuning Raises Uncertainty

Chat-V1 and Chat-V2 show **consistently higher SS** than base models across Llama-2, Yi, DeepSeek, and Falcon series.

Results – CP vs. Perplexity (InternLM-7B)

Table 2: Coverage Rate (%) & Set Size – target CR $\geq$ 90 %

	CR (%)		SS	
Task	**CP**	**PPL**	**CP**	**PPL**
QA	**90.68**	83.44	3.49	2.89
RC	**93.28**	95.48	2.19	2.39
CI	**90.10**	96.25	3.28	3.97
DRS	**90.40**	86.80	3.63	3.42
DS	**90.34**	87.13	4.47	4.33
Avg	**90.96**	89.82	3.41	3.40

PPL **fails** 90 % on QA, DRS, DS. CP **always meets** the coverage guarantee.

Results – ECE Comparison (InternLM-7B)

Table 3: Expected Calibration Error (%) – lower is better

Task	CP	Entropy	$P_{\max}$
QA	15.83	15.83	15.83
RC	**1.33**	1.32	1.41
CI	**3.16**	3.45	3.75
DRS	**12.11**	12.40	12.45
DS	**9.30**	9.30	9.62
Avg	**8.35**	8.46	8.61

CP achieves the **lowest average ECE** across all methods.

Our Experiment – Qwen3.5-4B on MMLU

Setup

- Model: `qwen3.5:4b` via Ollama (local inference)
- Dataset: MMLU ($n = 200$ test samples, 1-shot, base prompt)
- Logits via continuation logprob scoring (6 API calls/question)
- CP parameters: $\alpha = 0.1$, cal/test split 50 %

Results

Model	Acc (%) ↑	SS ↓	CR (%)
Qwen3.5-4B	53.40	3.96	**93.69**

CR $\geq$ 90 % target **met**. SS = 3.96 reflects moderate uncertainty on MMLU.

Conclusion & Future Work

Conclusion

- Accuracy alone is **insufficient** for LLM evaluation.
- Conformal prediction offers a **rigorous, model-agnostic** uncertainty measure.
- New **UAcc** metric jointly captures accuracy and certainty.
- Three findings: Acc $\neq$ certainty; larger $\neq$ more certain; finetuning raises uncertainty.

Future Work

- Extend CP to **free-form text generation** beyond MCQA.
- Apply to **multi-modal** and **closed-source** LLMs (GPT-4 etc.).
- Integrate uncertainty into **RAG** and **in-context learning** pipelines.